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$$O_k = \frac{1}{\psi(x;\theta)} \frac{\partial \psi(x;\theta)}{\partial \theta_k}$$

$$= \frac{\partial \log \psi}{\partial \theta_k}$$

Covariance of log-derivatives

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle$$

$$\Big(\mathrm{cov}(x_i, x_j) = \int dx_i\, dx_j\, (x_i - \langle x_i \rangle)$$

$$\times (x_j - \langle x_j \rangle) p(x_i, x_j)$$

$$E_L(x;\theta) = \frac{1}{\psi(x;\theta)} H(x)\,\psi(x;\theta)$$

$$\langle E_L \rangle$$

$$\vec{g}_i' = 2\left(\langle O_i' E_L \rangle - \langle E_L \rangle \langle O_i' \rangle\right)$$

$$\vec{\theta} \leftarrow \vec{\theta} - S_{ij}^{-1}\, \vec{g}_i'\, \delta\tau$$

Standard GD

$$\vec{\theta} \leftarrow \vec{\theta} - \eta\, \vec{g}_i$$

ADAM optimizer

Generative methods

$$D = \{x_1, x_2 \cdots x_N\}$$

$$x_i \sim P(D;\theta)$$

$$P(D;\theta) = \prod_{i=1}^{N} p(x_i;\theta)$$

$$\hat{\theta} = \arg\max_{\theta \in \mathbb{R}^p} P(D;\theta)$$

$$\log P(x;\theta) = \sum_{i=1}^{N} \log p(x_i;\theta)$$

$$\nabla_\theta \log P(x;\theta) = 0$$

score function

$$\nabla_\theta \log p(x;\theta) =$$

$$\frac{1}{p(x;\theta)} \frac{\partial p(x;\theta)}{\partial \theta}$$

$$p(x|\theta)$$

identity 1

$$\int_{x \in D} p(x;\theta)\, dx = \underline{1}$$

$$\frac{\partial}{\partial \theta} \int p(x;\theta)\, dx = \mathbb{E}_{x \sim p(x;\theta)} \left[\frac{\partial \log p}{\partial \theta} \right]$$

identity 2

$$\mathbb{E}_{x\sim p(x;\theta)}\left[\left(\frac{\partial \log p}{\partial \theta}\right)^2\right]$$

$$= -\mathbb{E}_{x\sim p(x;\theta)}\left[\frac{\partial^2 \log p}{\partial \theta^2}\right]$$

since the mean (identity 1) is zero

$$\frac{\partial \log p}{\partial \theta_k} = s_k$$

Fisher information matrix

$$F_{ij} = \mathbb{E}_{x \sim p(x;\theta)}\left[s_i s_j \right]$$

$$= \operatorname{cov}_{x \sim p(x;\theta)}(s_i, s_j)$$

$$= -\mathbb{E}_{x \sim p(x;\theta)}\left[\frac{\partial^2 \log p}{\partial \theta_i \partial \theta_j} \right]$$